

NEWBOROUGH WARREN

The Hidden Water Beneath the Dunes

Groundwater Decline, Conservation Management
and the Future of Protected Dune Slacks

A summary of hydrogeological research at Newborough Warren, Anglesey

For interested members of the public and conservation practitioners

M. Hollingham | March 2026

What this summary is about

Newborough Warren on the Isle of Anglesey is one of Wales's most important coastal nature reserves, protected for its rare and fragile dune slack communities — the shallow, seasonally flooded hollows between sand dunes that support nationally rare plants and insects. These habitats depend entirely on the groundwater beneath the dunes rising close to the surface each winter and staying close to the surface through summer.

Over more than 20 years, a network of 69 monitoring wells has recorded a steady decline in summer groundwater levels across the site. This document summarises the first comprehensive, data-driven analysis of why that is happening and what the evidence tells us about what may or may not help.

Where to find more

The full technical report, together with the analysis code, data, and supporting resources, is available at:

https://github.com/newbroman/Newborough_Hydrology

The landing page linked there provides access to the full report, an interactive scenario-modelling tool, and archived copies of all inputs and outputs.

1. The site: a dune system under pressure

Newborough Warren covers around 1,300 hectares of sand dunes at the south-western tip of Anglesey. It is designated as a Special Area of Conservation primarily for its dune slack communities — low-lying, seasonally waterlogged hollows where nationally rare plants such as creeping willow (*Salix repens*) and the liverwort *Petalophyllum ralfsii* have historically thrived.

The ecological value of these habitats depends on one hydrological condition: the groundwater table must stay close to the surface through summer, and rise close to or above the surface each winter. If summer water tables drop too far, the plant communities shift from wet slack species to dry slack species, and eventually to open dune. This transition is effectively irreversible on any management timescale.

The dunes sit on top of a body of groundwater stored in the sand. The eastern part of the site has shallow sand over impermeable glacial clay, producing a flashy, responsive water table. The western dunes have thick deep sand holding far more water but responding more slowly. The northern 700 hectares are covered by a Corsican pine plantation, planted between 1948 and 1965, running south from the bedrock ridge. Groundwater generally flows from the ridge either through the forest and out to the sea; or to Llyn Rhos Ddu the open dunes to the east and south which drain independently towards their own sections of coastline.

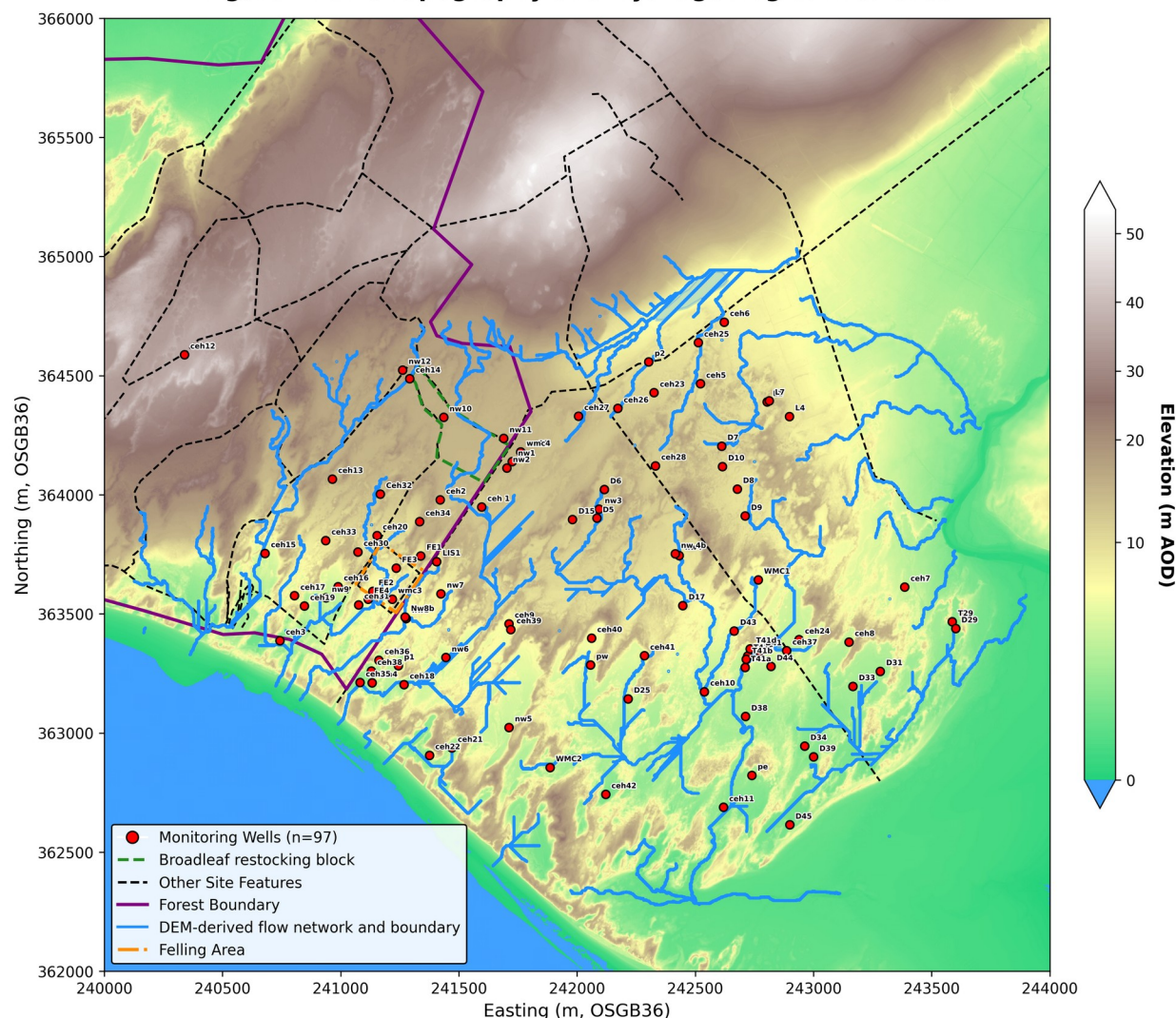
Figure 1: Site Topography and Hydrogeological Features

Figure 1. Site topography and hydrogeological features of Newborough Warren. The full monitoring network is shown on the digital elevation model, with the forest boundary (purple), the bedrock ridge along the northern edge, the DEM-derived drainage network (blue), and the 2017 felling area (orange dashed). The bedrock ridge forms the northern boundary of the study area; the coast lies to the south-west and south. Groundwater flows from the ridge along separate drainage pathways toward the sea.

The monitoring network

Groundwater levels have been measured monthly since 2005 using a network of dipwells — simple plastic pipes driven into the ground that allow the water table depth to be measured with a tape or contact meter. 69 wells have records long enough for the detailed analyses summarised here, with some extending back over 21 years. Rainfall and temperature data come from the Met Office weather station at RAF Valley, approximately 16 km to the north.

“The ecological window for effective intervention is narrow and closing. Western-block water tables have already fallen beyond the point at which any realistic winter rainfall can restore surface flooding.”

2. What the data shows: a system in decline

The most important finding from 21 years of monitoring is straightforward: summer groundwater levels are falling, and they are falling across every part of the site.

Every structural zone of the site — the plantation, the western dunes, and the eastern dunes — is showing a statistically significant decline in the depth to which the water table falls each summer.

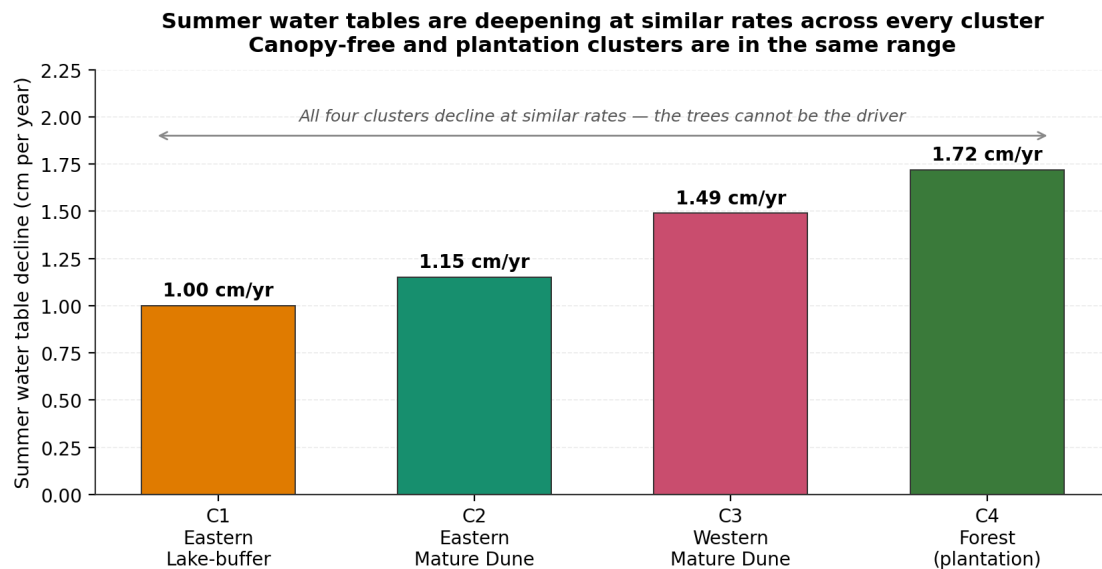


Figure 2. Annual rate of decline in summer minimum water table depth across the site's four principal hydrogeological zones. The Forest cluster (C4) shows the steepest decline, consistent with forest water use drawing down the water table under and directly downstream of the plantation. The two Eastern Block clusters (C1, C2) and the Western Block (C3) — which drain independently of the forest — are nevertheless each declining at around 1 to 1.5 cm per year.

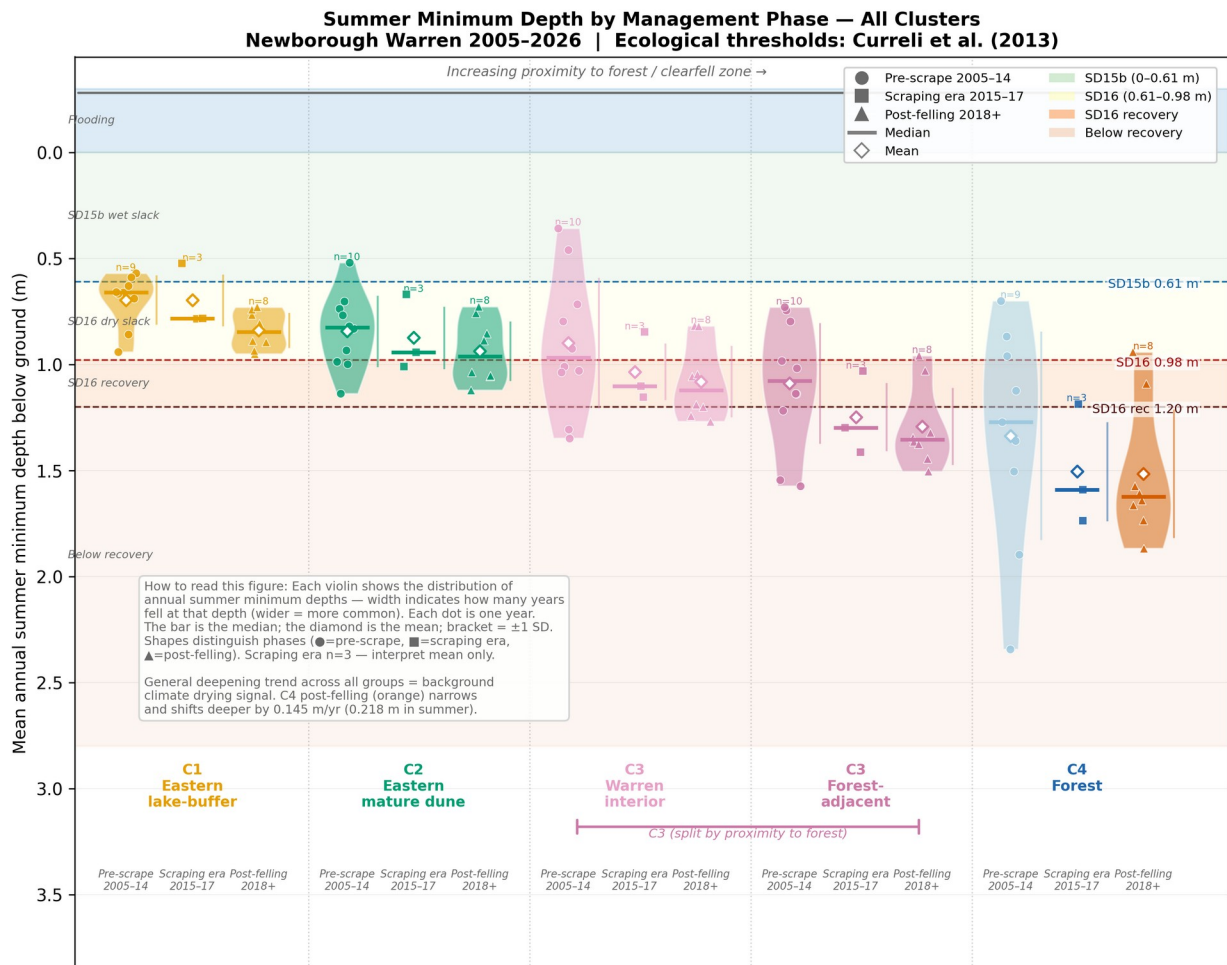


Figure 3. Summer minimum water tables across three management eras, cluster by cluster. Each violin is a distribution: wider means more years fell at that depth. The three eras are the pre-scraping period (2005–14), the period between the April 2015 scraping at CEH36 and the December 2017 clearfell (2015–17), and the post-felling period (2018+). Every cluster — including C1 (Eastern lake-buffer), which is on a separate drainage system and unaffected by either intervention — has shifted deeper from era to era. The C3 Western Block is split here into warren interior wells (hydrologically independent of the forest) and forest-adjacent wells (downstream of the plantation); forest-adjacent wells sit consistently deeper than the warren interior, consistent with the forest drawing the water table down along its own drainage pathway. C4 Forest narrows and shifts deeper post-felling.

Projected Climate Trajectory vs Ecological/Flooding Thresholds: Newborough Warren

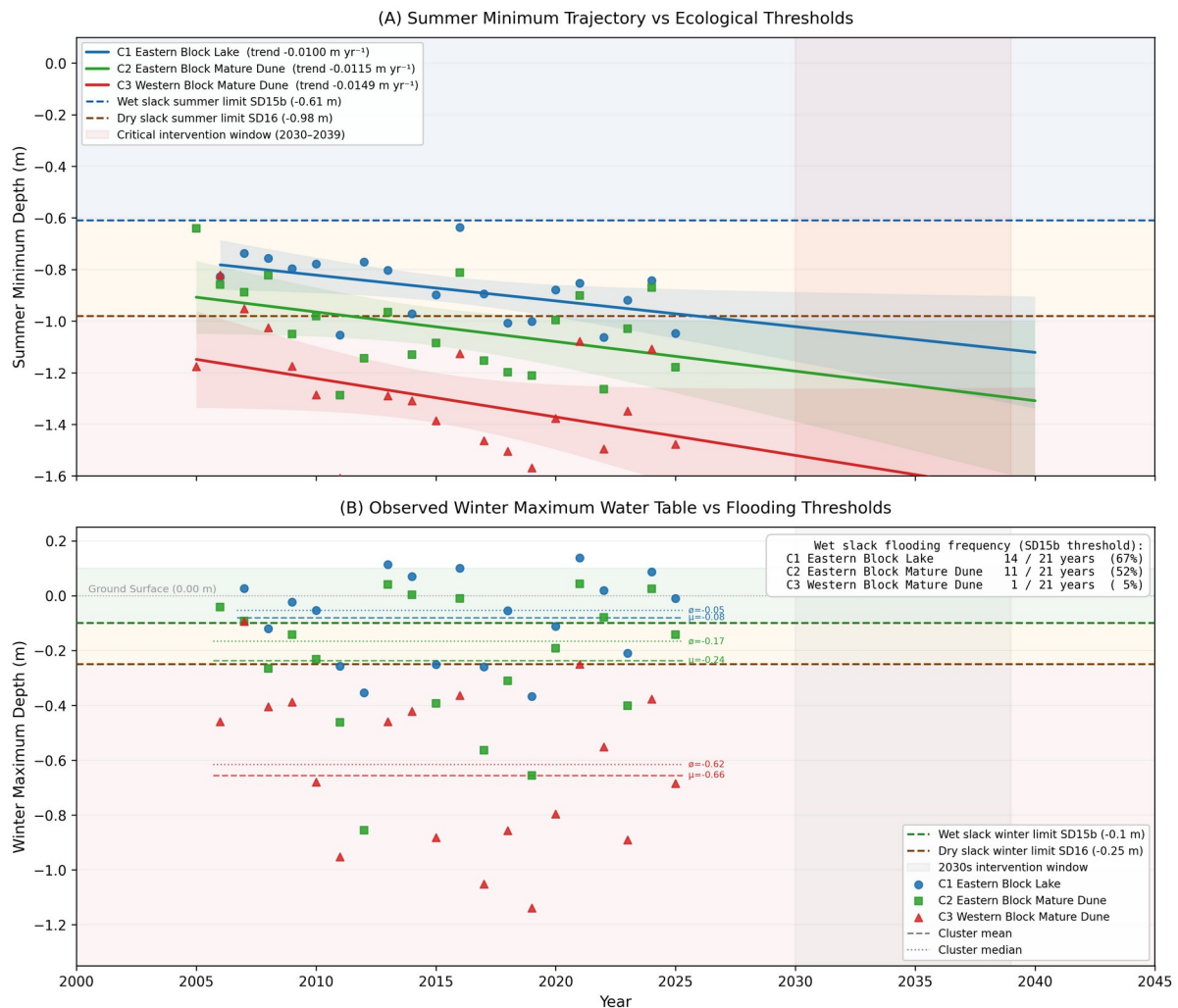


Figure 4. Summer minimum (top) and winter maximum (bottom) water tables against ecological thresholds, 2005–2025 with projections to 2040. Summer minima are deepening in every monitored cluster; winter maxima show no significant trend. The Western Block (red) has been below the wet-slack flooding threshold in 20 of 21 monitored years.

Why does this matter for nature?

Research by Curreli and colleagues (2013) established that the entire gradient from wet slack plant communities to dry slack communities is compressed into a water table range of less than 40 cm. A shift of just 15 cm in summer water table depth can determine whether a slack supports rare wet-loving plants or becomes dry scrub. The decline rates recorded here — 1 to 1.7 cm per year — mean that by the early 2030s, even the eastern slacks that currently flood reliably each winter may begin crossing into ecologically critical territory.

At the same time, summer temperatures have been rising. The RAF Valley record shows a statistically significant warming trend over the past 95 years. Hotter summers draw more water out of the ground. 2022 was particularly severe — hot and dry — with summer temperatures more than 1.6 °C above the long-term average.

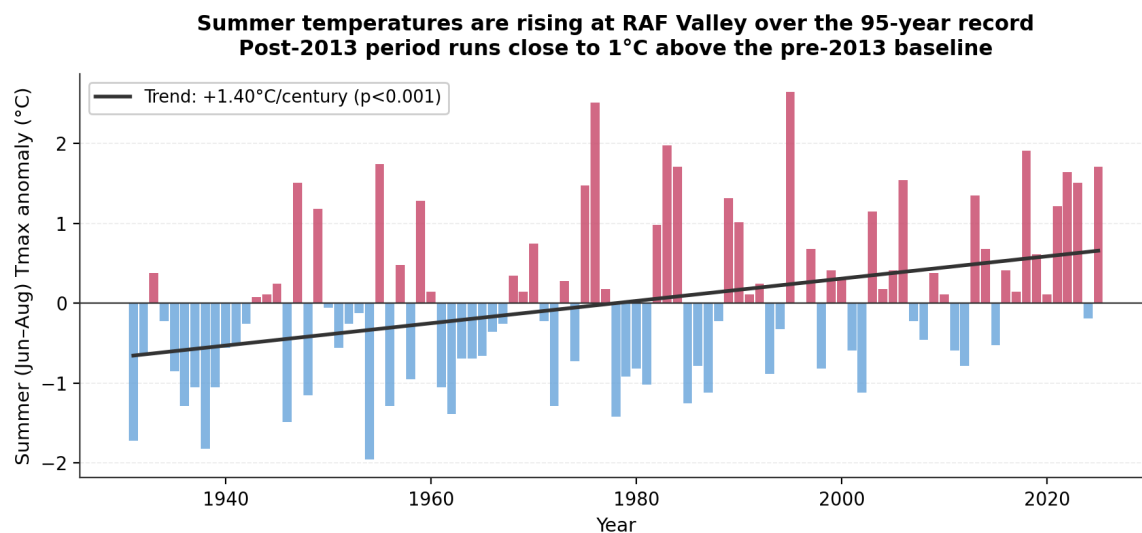


Figure 5. Summer temperature anomaly at RAF Valley (Met Office), 1930–2025. Red bars indicate summers warmer than the 95-year mean; blue bars indicate cooler summers. The trend line shows a statistically significant long-term warming of approximately 1.4 °C per century ($p<0.001$), with the post-2013 period running close to 1 °C above the pre-2013 baseline.

3. Understanding the aquifer: three types of behaviour

Not all parts of the site behave the same way, and understanding why matters for interpreting any management response. Statistical analysis of the 21-year well record identified three fundamentally different groundwater behaviours across the site.

The shallow pan (Eastern Block)

Where thin sand sits over glacial clay, there is very little storage depth for rainfall to fill before the water table rises — so it responds quickly and sharply to each rainfall event. But the same thin geometry means water drains away quickly once the rain stops, moving toward the coast. The water table is flashy in both directions: fast to rise, fast to fall. These slacks flood most reliably (67 per cent of years in the lake-buffer zone) because the water table needs only a modest rainfall event to reach the surface — but the flooding can be short-lived if dry conditions follow.

The deep sponge (Western Block)

Where thick, deep sand extends across the western part of the site, the aquifer acts like a giant sponge — slowly absorbing rain over a large unsaturated depth before the water table responds. These slacks can hold water longer but need far more rainfall to raise the water table to the surface. The western slacks currently flood in only 1 in 21 monitored years, and at present summer depths flooding is essentially impossible regardless of how much rain falls in winter.

The canopy sponge (Forest)

The Corsican pine plantation intercepts approximately 24 per cent of incoming rainfall in its canopy before it reaches the ground, suppressing recharge. The water table beneath the forest is correspondingly deeper than in the surrounding open dunes. The water table beneath the forest is also measurably less sensitive to summer evaporative demand than the water table under open dune — see Section 4 for the evidence base and interpretation.

4. The plantation: what the evidence shows about its hydrological role

For decades, the Corsican pine plantation has been identified as a major cause of water table decline at Newborough Warren, on the grounds that it intercepts rainfall and uses water through transpiration. The monitoring evidence now allows this assumption to be tested quantitatively, and the picture that emerges is more nuanced than it first appeared.

The plantation does reduce recharge — this is confirmed. It also appears to draw the water table down beneath its own footprint and along its drainage pathway toward the sea. What the evidence does not support is the proposition that the plantation is responsible for water table decline across the whole site. Summer decline is happening in clusters that have no trees and are on separate drainage systems from the forest (see Section 8).

The plantation changes how the water table responds to summer heat

Each cluster of monitoring wells was fitted with a statistical model that quantifies how sensitive the water table is to monthly evaporative demand. Two numbers drop out of this analysis and are directly comparable across clusters:

- For the open-dune clusters, the water table falls by roughly 4–5 mm for every mm of potential evaporation in an average month.
- For the Forest cluster, the water table falls by roughly 3.4 mm per mm of potential evaporation — measurably less sensitive than the open dunes.

One plausible reading of this difference is that the canopy intercepts a portion of the evaporative demand at the surface before it can reach down to the water table, acting as a sacrificial evaporative layer. An alternative reading is that the canopy simply changes the hydraulic environment below ground in some way that reduces the PET-to-water-table coupling. A separate analysis in the full report tested whether the forest's low sensitivity to PET might instead reflect the water table being too deep for roots to reach (capillary disconnection) and found no evidence for this: the Forest cluster's low PET sensitivity is a surface canopy effect rather than a depth effect.

It is worth being explicit about what this finding does and does not say. The monitoring evidence is consistent with the water table beneath the plantation being less sensitive to summer evaporative demand than the water table under open dune. It does not rule out the plantation drawing the absolute water table deeper than open dune would in the same location — canopy interception removes roughly a quarter of incoming rainfall, and the plantation's transpiration demand is real. These two statements can be true simultaneously: the plantation may both draw the water table deeper and make it less reactive to weather swings.

The evidence on broadleaf conversion

Conversion of the plantation to broadleaf woodland is sometimes proposed as an alternative to Corsican pine management. The monitoring data from a small 1996 broadleaf restocking block

allow this question to be examined, though the evidence comes from a single monitored dipwell (NW10) and must be treated as indicative rather than conclusive.

The broadleaf block appears to produce deeper summer minima than adjacent Corsican pine at comparable elevation. If real, the implied mechanism is two-phase rather than single. During the deciduous dormant phase (autumn and winter), reduced canopy interception allows greater throughfall to the aquifer, producing more recharge but also more drainage. During the leaf-out phase (late spring and summer), the larger summer leaf area increases evapotranspiration demand. The summer minimum is then deepened by both halves: more pre-summer drainage leaving less storage available, plus a higher summer draw drawing it down further. The signal shows up as a larger seasonal amplitude rather than as simply higher summer evaporation.

The monitoring evidence therefore suggests broadleaf conversion may be unhelpful for dune slack hydrology, but a single dipwell cannot settle the question. It is flagged here as a hypothesis for further investigation rather than a settled conclusion.

Why the picture has changed since earlier studies

Earlier studies linking water table decline to the plantation were conducted during the driest decade in the 95-year climate record, when annual rainfall averaged around 793 mm against a long-run mean of 856 mm. The statistical tools now available to separate climate effects from land cover effects were not then in use. With a longer record and methods that can partition the two, the current study finds that while the forest exerts a real local hydrological effect, the site-wide decline has another source.

5. The December 2017 clearfell: what the monitoring shows

In December 2017, approximately 8.4 hectares of plantation — just over 1 per cent of the total — were felled at the plantation–dune boundary. The felling was itself set up as a management experiment: a deliberate small-scale intervention intended to observe the hydrological consequences of canopy removal on an open-dune slack system. This analysis extends that experiment by applying statistical methods that can separate the intervention’s effect from the climate signal present over the same period — something that was not feasible when the experiment was set up but becomes possible with eight years of post-felling data and a longer reference record.

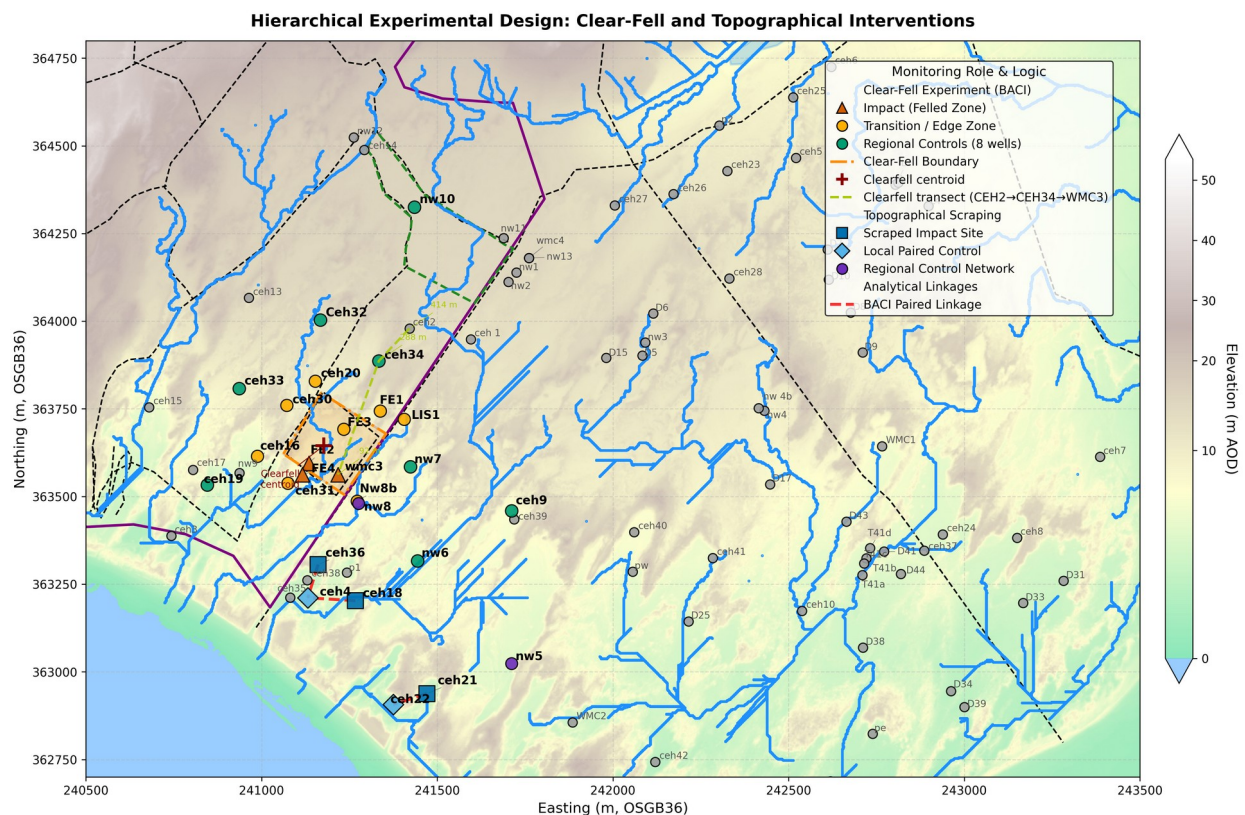
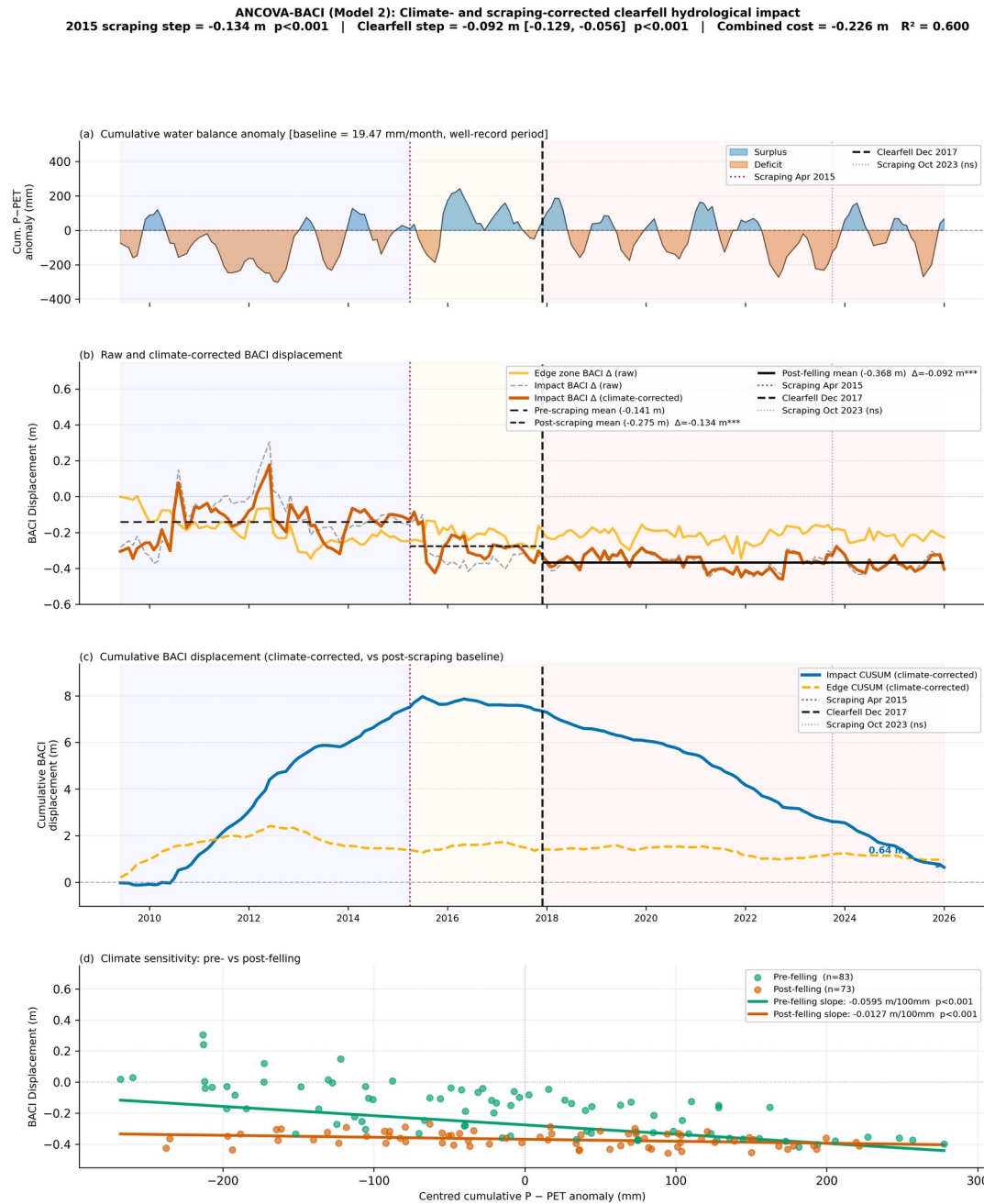


Figure 6. The clearfell experimental layout. Core impact wells (red triangles) sit within or immediately adjacent to the felled block. Transition or edge-zone wells (orange circles) lie along the broader forest–dune boundary. Regional control wells (green circles) lie outside the zone of direct influence. Scraped sites (blue squares) at CEH36, CEH18 and CEH21 are also shown.

After correcting for climate variability, the 2017 clearfell caused a step lowering of approximately 9 cm in the core impact zone, on top of an 11 cm lowering attributable to the earlier April 2015 scraping at CEH36, which sits downslope of the clearfell area and had already been drawing the upslope water table down (see Section 6). The combined hydrological cost was around 21 cm of water table depression relative to the pre-intervention baseline. By early 2026, eight years after the felling, the accumulated deficit showed no sign of stabilising.



† Oct 2023 scraping term: $\text{coef} = -0.031\text{ m}$, $p = 0.258$, $\Delta\text{AIC} = +0.66$ — not retained in final model.

Figure 7. Climate-corrected analysis of the December 2017 clearfell. Panel (a) shows the cumulative wet/dry anomaly (rainfall minus evaporative demand, relative to the long-term mean) over the monitoring record — the index used to partition climate variability from intervention effects. Panel (b) shows the water table displacement in the impact zone relative to controls: raw (dashed grey), climate-corrected (solid orange), and the edge zone for comparison (yellow). The step-downs at the 2015 scraping and 2017 clearfell events are marked. Panel (c) shows the cumulative climate-corrected displacement, which continues to decline after felling rather than stabilising. Panel (d) shows a change in the relationship between impact-zone displacement and the wet/dry anomaly before and after felling; the change is statistically clear but its mechanistic interpretation is complex and discussed in the full report rather than treated as a conclusive finding here.

What the clearfell appears to have changed

The evidence in Figure 7 points to three things, each with a different strength of support:

- The water table in the impact zone stepped down by roughly 9 cm at the clearfell event (after removing the climate signal), on top of an 11 cm step attributable to the upslope-draining effect of the 2015 scraping at CEH36 (see Section 6). This is the combined hydrological cost shown at the top of Figure 7.
- Effective recharge from rainfall increased. The roughly 24 per cent of incoming rainfall that had been intercepted by the canopy now reaches the ground. This is consistent with expectations.
- The way the impact zone responded to long-term wet and dry conditions changed relative to the open-dune controls. Panel (d) of Figure 7 illustrates this. Interpreting this change mechanistically is not straightforward, and the full report treats it as a statistical observation rather than a piece of mechanism-level evidence.

The statistical model used for this analysis is a form of ANCOVA-BACI — a before-and-after comparison that also includes a running index of cumulative wet-versus-dry conditions, so that the effect of the felling can be separated from the effect of the weather. The full equation, fitted values and robustness tests are set out in the main report. For the purposes of this summary, what matters is the size of the step change at the felling event (panels a–c) and the persistence of the deficit that follows it.

What this means

The clearfell increased the amount of rainfall reaching the aquifer, as expected. But the impact-zone water table also stepped down at the felling event and never recovered — by early 2026 the cumulative deficit was still accumulating. Because summer minimum depth is the variable that matters most for dune slack ecology — and, separately, the dominant control on whether winter flooding occurs in the Western Block — the net effect on the ecological target was negative.

6. Dune scraping: a real but not hydrologically neutral tool

Dune scraping involves physically removing a layer of sand from a slack floor to bring the ground surface closer to the water table. It is a direct mechanical intervention that does not depend on changing hydrological processes — it simply reduces the gap the water needs to bridge.

Scraping at CEH36 (April 2015) delivered a measurable climate-corrected net benefit of +14 cm during the period before the 2017 clearfell — roughly one third of the entire hydrological gradient that separates a dry slack from a wet slack. At the local scale this is a genuinely meaningful outcome: lowering the slack floor brought the ground closer to the water table, as intended.

There is, however, a subtler consequence of the intervention. A scraped hollow acts as a drainage sink for the area upslope of it — water that would otherwise have sustained the upslope water table now has somewhere lower to drain to. At CEH36, the area upslope of the scrape includes the plantation block that was subsequently felled in December 2017. The scraping therefore contributed to lowering the water table across the area later studied in the clearfell BACI experiment, independently of the felling itself. This is why the statistical analysis of the clearfell is careful to separate the scraping step (approximately –14 cm at the impact zone) from the felling step (approximately –9 cm) — the two interventions were not hydrologically independent.

A separate scraping intervention at CEH18, further along the coast and hydrologically unrelated to the clearfell area, did not deliver a detectable climate-corrected net benefit. The two sites differ in their position within the drainage system and in the character of the local aquifer; one case of success and one case of no detectable benefit is not enough to generalise about where scraping works and where it does not. What the evidence does suggest is that scraping should be understood as a local intervention with non-local consequences: useful for making the scraped slack wetter, but also redistributing water from the area upslope of it.

7. Forecasting next summer's water table

The single most important variable for dune slack ecology is summer minimum water table depth — how close the water table stays to the surface through summer. It is also, separately, the dominant control on whether slacks flood the following winter. The research provides a simple statistical tool for forecasting it.

For each structural domain of the site, an equation predicts next summer's minimum water table depth ($h_{\min}$) from two pieces of information that managers routinely have: this winter's maximum water table depth ($h_{\max_winter}$, measured in dipwells) and total summer rainfall (P_{summer} , from the nearest climate station):

$$h_{\min} = b \cdot P_{\text{summer}} + c \cdot h_{\max_winter} + \text{constant}$$

The fitted equations for each structural domain are:

Structural domain	Equation	R ²
Eastern Block (C1, C2)	$h_{\min} = 0.00093 \cdot P_{\text{summer}} + 0.333 \cdot h_{\max_winter} - 1.183$	0.41
Western Block (C3)	$h_{\min} = 0.00096 \cdot P_{\text{summer}} + 0.577 \cdot h_{\max_winter} - 1.214$	0.77
Forest (C4)	$h_{\min} = 0.00079 \cdot P_{\text{summer}} + 0.717 \cdot h_{\max_winter} - 1.180$	0.92

Table 1. Summer drought transfer functions for each of the three structural domains of Newborough Warren (Hollingham 2026, Table 9). P_{summer} is total April–September rainfall in millimetres; $h_{\max_winter}$ and $h_{\min}$ are in metres, expressed as negative numbers below the surface. R^2 is the proportion of year-to-year variation in summer minimum depth that the equation explains.

These equations explain 41%, 77% and 92% of the year-to-year variation in summer minimum depth in the Eastern Block, Western Block and Forest respectively. They are notably more reliable than the equivalent winter-flooding equations, because summer minimum depth is strongly tied to where the water table started the season.

A worked example: Western dune slack

Take a Western Block slack where last winter's maximum was 0.30 m below the surface, and 200 mm of rain fell between April and September.

$$h_{\min} = 0.00096 \times 200 + 0.577 \times (-0.30) - 1.214 = -1.196 \text{ m}$$

The summer minimum would be expected around 1.2 m below the surface — deep into dry-slack territory. For the same slack to return a summer minimum shallower than the wet-slack threshold of 0.61 m, either last winter would have had to peak close to the surface or summer rainfall would need to exceed 600 mm. Neither is realistic in a typical year.

What this reframes

Managers cannot reliably forecast slack flooding from seasonal rainfall forecasts alone. What matters is how deep the aquifer drained the previous summer, and how close the winter maximum comes to the surface. Summer dipwell readings are therefore the most valuable diagnostic measurement on the site. Interventions that prevent the water table from draining deeply in summer — or that hold the winter maximum closer to the surface — are more valuable than interventions that try to add more water in winter.

The full report also contains the reverse calculation — the P_{flood} equations (Section 3.6.3 and Table 10 of the main report) give the minimum winter rainfall required to raise the water table to the slack floor from a given summer starting position. For Western Block slacks, the honest answer is usually “more rainfall than actually falls in a winter”. The summer transfer functions above reach the same conclusion from the other direction.

The companion modelling suite

The cluster-level state-space model described throughout this report has been implemented in a companion browser-based scenario viewer, the Newborough Warren Hydrogeological Modelling Suite:

https://newbroman.github.io/Newborough-Hydrology_models/scenario_viewer.html

The viewer lets users explore scenario simulations of climate change, clearfell extent, and scraping interventions under the fitted state-space framework. It is a research prototype under development, and should be treated as indicative rather than as an operational forecasting tool pending field validation.

The viewer does not itself calculate the forecasting equations in Table 1. To apply them to specific sites, paste the relevant formula into a spreadsheet and supply the inputs from current dipwell and rainfall data.

8. What appears to be driving the decline

Why not the forest?

A reasonable first response to the decline rates in Figure 2 is: "C4 (Forest) is dropping fastest at 1.7 cm per year; isn't that a clear sign the plantation is the cause?" It would be, if the Forest cluster were the only one declining. It is not.

The three open-dune clusters — C1 (Eastern Block Lake-buffer), C2 (Eastern Block Mature Dune) and C3 (Western Block Mature Dune) — are each declining as well, at roughly 1.0, 1.15 and 1.5 cm per year respectively. The important point is not how far they are from the forest in raw distance (most of them are within a kilometre) but how the water actually flows at the site. The plantation sits along a drainage pathway that runs from the bedrock ridge to the north, through the forest, and out to the sea. C1 and C2 — the Eastern Block clusters — sit on a different drainage system, running eastward and southward to their own section of coastline. Most of C3 drains westward and southward independently of the forest. There are C3 wells that sit below the main plantation on its south-bound drainage pathway (the "forest-adjacent" wells in Figure 3), and those can reasonably be influenced by forest water use; but the rest of C3, and all of C1 and C2, are hydrologically independent of the forest.

For the forest to be the site-wide driver, it would therefore need to draw water out of clusters that are on entirely different drainage systems. There is no hydrological pathway by which that would work. Figure 3 shows this directly: every cluster, including C1 (which is on a fundamentally separate drainage system from the forest), has shifted deeper from era to era. That progression cannot be explained by the forest.

The more parsimonious reading is that the forest is a real local driver — acting on its own footprint and its immediate drainage pathway — but that something else, operating across the whole site, is responsible for the progression visible in every cluster. The forest's additional rate on top of the open-dune rate (the 0.2 to 0.7 cm/yr gap) is consistent with forest water use adding to the common signal in the clusters it can reach.

"C1 sits on an entirely different drainage system from the forest. The forest cannot reach it. And yet C1 has shifted deeper in every successive era. Whatever is doing that, it is not the forest."

Two candidate site-wide drivers

The monitoring evidence does not definitively pin down what that "something else" is. The most direct candidate is **rising summer evaporative demand**: hotter summers pull more water out of the ground, and the RAF Valley record shows summer maximum temperatures rising at approximately 1.3–1.4 °C per century, with the post-2013 period running close to 1 °C above the pre-2013 baseline (Figure 5). Winter rains then have to work harder to restore the water table, and a ratchet effect accumulates: each summer pulls down a little further, and the recovery does not quite catch up. Over 21 years, a small annual imbalance becomes the 1–1.5 cm/yr decline we observe.

The water balance analysis (Figure 8 below) also shows a **residual flux** in every cluster — a mass-balance gap that direct rainfall recharge does not close. Something is sustaining the water table beyond what direct rainfall can account for, and that "something" differs across the site: lake seepage at C1 (where Llyn Rhos-Ddu intercepts and stores ridge recharge), ridge recharge arriving directly at C4, and a mix of contributions at C2 and C3. Whether any of these lateral sources has itself been declining — which would add to the climate driver — is not something the monitoring data alone can resolve, and requires direct measurement at each source candidate (see Section 11).

Both explanations share a common upstream driver: a warmer, drier summer climate. Rising summer temperatures directly increase evaporative demand, and — if they also reduce the moisture reaching the ridge recharge zone over the longer run — reduce the lateral inflow that sustains the dune field. Of the two, the evidence for rising evaporative demand is more direct (Figure 5). The evidence for declining ridge subsidy is circumstantial; the residual that ridge subsidy is inferred to close is real and well established, but separating a long-run decline in ridge subsidy from other long-run drivers would require measurements on the ridge itself (see Section 11).

Why the aquifer is vulnerable to either driver

There is also a structural feature of this particular aquifer that makes either driver hard to absorb. Sand drains rapidly — water stored over winter flows away quickly toward the sea. This means a wetter winter does not necessarily translate into a higher summer water table, because the extra recharge has mostly drained away before summer begins. The aquifer is predisposed to summer stress regardless of which of the two candidate drivers proves dominant.

8.1 The water balance residual

The research identifies a substantial water balance residual in every cluster — a flux required to close the monthly water balance that direct rainfall recharge does not explain. The residual is real (confirmed by mass balance) and spatially structured, but the pattern differs from cluster to cluster. The largest residual is at C1 (Eastern Lake-buffer), where Llyn Rhos-Ddu sits immediately adjacent to the cluster. The lake intercepts any ridge recharge that would otherwise have flowed directly to C1, integrates that water with direct rainfall on the lake and runoff from the surrounding catchment, and releases it gradually through seepage into the C1 aquifer. The lake is simultaneously a ridge-recharge trap and a buffering store, and C1's water table is sustained by what leaks out of it. At C4 (Forest) the residual is geometrically consistent with lateral inflow from the ridge arriving directly through the plantation. At C2 and C3, smaller residuals are plausibly a mix of ridge inflow through the warren strip, cross-site drainage, and model structural effects. Two diagnostic tests run on the monitoring data (a pulse-delivery travel-time test and a summer-evaporation test) both returned null results — the residual is compatible with several of these mechanisms but none of them can be uniquely confirmed from water-level data alone.

It is worth being explicit that the ridge does not feed the site through a single pathway. Three separate pathways can be distinguished from the monitoring network, each feeding different parts of the dune field:

- **Ridge → forest → sea.** This is the pathway feeding the Forest cluster (C4) and the forest-adjacent C3 wells downstream. It is the pathway for which the evidence is strongest: an independent boundary-intercept audit identifies ridge-derived lateral recharge as the dominant sustaining mechanism for C4, with the largest single boundary subsidy in the network measured at CEH14 ($\alpha \approx +0.21$ m/month). The ridge slopes along this pathway are fully covered by the Corsican pine plantation.
- **Ridge → Llyn Rhos-Ddu → C1 lake-buffer.** The lake sits between the ridge and the C1 wells, and effectively intercepts the ridge-to-C1 pathway. What C1 wells read is the state of the lake-fed downstream system, not the ridge directly. No strong direct ridge-subsidy signal is detected at C1 for this reason.
- **Ridge → open warren → C2 and warren-interior C3.** This pathway reaches the open dunes mostly without passing through the forest. A narrow strip of open ridge approximately 300 m wide lies between the plantation and the lake, but its surface drainage runs eastward into the lake rather than southward toward the warren. The effective uncovered pathway is therefore smaller than the map suggests. The monitoring network does not include wells close enough to the ridge north of C2 and warren-interior C3 to confirm the lateral inflow directly, but the water balance deficit in these clusters is consistent with a similar mechanism.

Two diagnostic tests have been run to check whether specific mechanistic attributions of the residual can be confirmed from the monitoring record. A time-lag test checked whether rainfall signals propagate through the ridge aquifer to the dune field with a travel time, as would be expected if ridge recharge arrives as seasonal pulses — it did not. A seasonality test checked whether the residual has the summer-heavy negative signature that unmodelled evaporation would produce — it did not. Both null results are informative: the residual is compatible with ridge recharge, but only in a steady-baseflow form indistinguishable from a constant offset; it is equally compatible with model-structural effects that the linear state-space framework does not capture. Without direct measurement on the ridge itself, these alternatives cannot be separated.

Would clearfelling the forest release more ridge subsidy to the dune field?

A reader familiar with the site may reasonably ask: if the ridge-to-C4 pathway runs through forested ridge slopes, and if the forest is intercepting and consuming some of the ridge water before it reaches the dune slacks below, would clearing that part of the forest not release more ridge subsidy to the dune field?

This is a serious argument, and the 2017 clearfell was in effect a partial test of it: 8.4 hectares of forest were felled on the ridge-to-sea pathway. The outcome at the downstream slacks was not wetter conditions. The impact-zone water table stepped down approximately 9 cm at felling and the deficit has continued to grow for eight years (Figure 7). Whatever interception the

canopy was removing from the ridge pathway, the net effect of removing it was drier slacks below, not wetter.

Two caveats are worth noting alongside that test. First, the 2017 felling was a small area on a specific part of the ridge drainage pathway; it is not a clean test of what wider or differently designed clearance might do. Second, the Corsican pine canopy is relatively light compared with denser evergreen species (Sitka spruce, for example, intercepts considerably more rainfall for the same stocking density), so the interception component of the ridge-to-forest pathway is not as severe as under some alternative species. The evidence argues against large-scale clearance as a route to increasing ridge subsidy, but it does not exhaust the space of canopy management options the ridge pathway might respond to.

Two broader observations also follow. First, even if clearfelling the forest had increased subsidy to the ridge-through-forest pathway, it would not address the decline observed at clusters on the other two pathways (C1 lake-buffer, C2 Eastern Mature Dune, warren-interior C3). The ridge pathways are distinct, and what affects one does not automatically affect the others. Second, actions that might reduce lateral recharge at its origin — for example, forestry, drainage or abstraction on the ridge slopes themselves, north of the study area — would have consequences for all three pathways simultaneously. The hydrology of the dune field appears to depend on the integrity of what happens upslope of it.

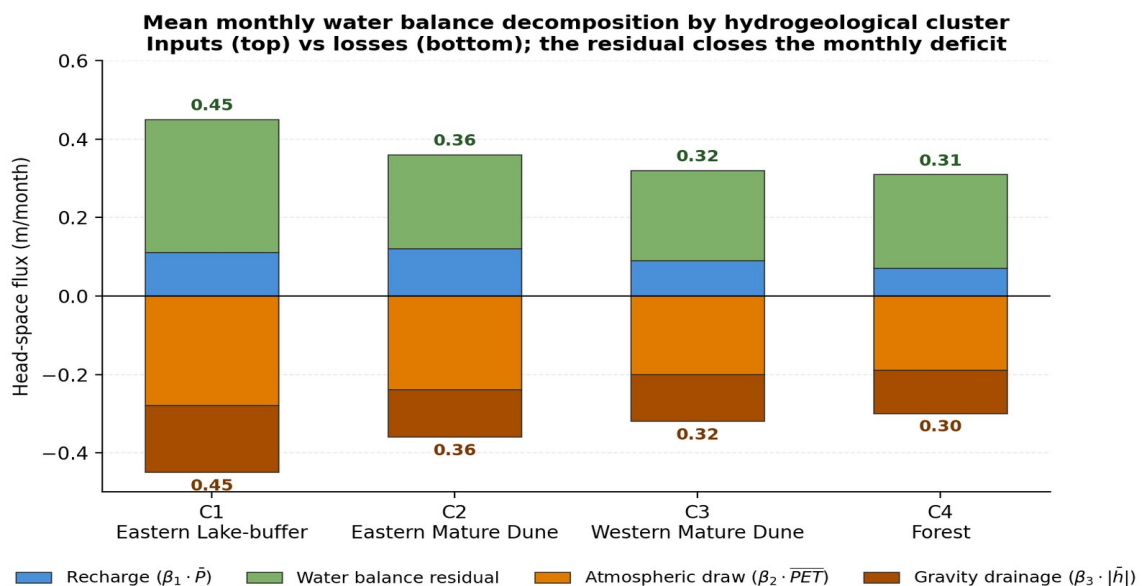


Figure 8. Mean monthly water balance by cluster. Local rainfall recharge (blue) is consistently smaller than losses (orange + dark brown) in every cluster. The water balance residual (green) is the flux required to close the monthly deficit. The residual is largest at C1 (where Llyn Rhos-Ddu acts as a local store of intercepted ridge recharge and releases water to the aquifer gradually) and at C4 (where direct lateral inflow from the ridge through the plantation is geometrically plausible). Values are expressed in head-space units (metres of water table change per month), the natural unit of the state-space model.

9. What the findings suggest

The monitoring evidence points to a set of observations and questions rather than a list of prescriptions. Decisions about how to manage the site rest with those responsible for it, and will need to balance the hydrological findings reported here against ecological, landscape, policy, biodiversity and operational considerations that are outside the scope of this work. What follows is offered in that spirit.

The ecological variable that matters most is summer minimum depth

Dune slack communities depend on shallow water tables through summer. Summer minima also turn out to be the dominant control on whether winter flooding occurs in the Western Block, rather than winter rainfall amount. Diagnostic monitoring and interventions are therefore most usefully focused on summer, not winter.

The decline is widespread, not confined to the forest

Summer water tables are falling in every cluster, including clusters on drainage systems independent of the plantation. The forest cannot be the site-wide driver. The residual site-wide signal aligns with independently observed summer warming at RAF Valley. The possibility of a long-run decline in ridge subsidy cannot be ruled out but is less directly supported by the evidence at present. Site management cannot by itself reverse the underlying trajectory; it can influence how steeply individual slacks decline and how resilient they are to it.

The evidence on large-scale clearance is mixed

The 2017 clearfell increased recharge to the aquifer as expected, but the impact zone also stepped down at felling and has continued to accumulate a deficit relative to the controls ever since. In net terms, the monitoring evidence suggests clearance as implemented in 2017 has been unhelpful for dune slack hydrology in the area that was felled. Whether this generalises to other plantation blocks, other scales of clearance, or other intervention designs is an open question.

Scraping works locally, but is not hydrologically neutral

The 14 cm climate-corrected net benefit recorded at CEH36 during the pure scraping era suggests scraping can deliver meaningful hydrological improvement at the scraped location itself. The mechanism is simple: lowering the slack floor brings the ground closer to the water table. What the subsequent clearfell analysis makes clear is that a scraped hollow is also a drainage sink for the area upslope of it — a property that is useful for making the scraped slack wetter, but that propagates a water table drawdown into the surrounding area.

The water balance has a substantial residual whose components are not fully resolved

The water balance analysis shows a structured residual in every cluster that direct rainfall recharge does not account for. The character of the residual differs from cluster to cluster: at

C1 the most likely source is seepage from Llyn Rhos-Ddu, which both intercepts ridge recharge arriving from the north and stores water for gradual release; at C4 the most likely source is lateral inflow from the ridge arriving directly through the plantation; at C2 and C3 the residual is smaller and likely reflects a mix of contributions. Actions that might affect any of these — changes to the water balance of Llyn Rhos-Ddu; forestry, drainage or abstraction on the ridge slopes to the north; or interruption of cross-site drainage pathways — could have consequences that would not be visible in the dipwell network until they had been operating for some time. Direct measurement at the source candidates (flagged in Section 11) would be the most informative next step.

The ecological intervention window is narrow

Projection of current trends suggests that eastern dune slacks — currently the most ecologically viable part of the site — will begin crossing critical threshold water table depths in the early 2030s. This narrows the window during which restoration or maintenance interventions can realistically deliver ecological value. Western slacks may already be outside that window for the level of summer deficit they have accumulated.

What the research does not settle

Several important questions remain genuinely open. The broadleaf conversion hypothesis is based on a single monitored well and requires replication. The net hydrological balance of canopy management at different intensities (thinning, partial felling, patch clearance) cannot be inferred from a single 2017 clearfell event. The spatial extent to which clearfell-driven changes have propagated into the wider dune system cannot be quantified from the current monitoring network. Whether the ridge recharge interpretation is robust across all clusters, or applies most strongly to the forest margin, depends on field measurements that have not yet been made. These are avenues for further work, not decided matters.

10. The bigger picture: adaptation, not just restoration

Perhaps the most important message from this research is a shift in how the management challenge may need to be framed. For many years, the goal at Newborough Warren has been described as restoration — returning the water table to a historical baseline. The evidence summarised here suggests that framing may no longer be adequate.

The site-wide component of the water table decline appears to be climate-driven. The most direct evidence points to rising summer evaporative demand; a long-run decline in ridge subsidy cannot be ruled out as a contributing factor but is less directly established. A management regime that could reverse the underlying trajectory through land-cover change alone is not apparent in the evidence. A more realistic framing may be adaptation: maintaining the best possible hydrological conditions under an increasingly challenging climate, for as long as the ecological window remains open, and planning deliberately for the transitions that will occur as that window closes.

This means acknowledging that western dune slacks are already in a structurally compromised state from which full restoration may not be achievable under current climate conditions. It means concentrating monitoring and care on the eastern slacks, which still flood reliably, and managing the transition carefully to delay their deterioration. And it means using the forecasting framework developed in this research to track the trajectory year by year and respond as evidence accumulates.

A note on broadleaf conversion

Some proposals for the Newborough plantation have included conversion to broadleaf woodland. The monitoring evidence from a single broadleaf block at the site suggests this may not be beneficial for dune slack hydrology: broadleaf trees retain summer transpiration demand while losing their leaves in autumn, precisely when the aquifer most needs to accumulate water.

This is one dipwell's worth of evidence — indicative rather than conclusive — and the question of broadleaf conversion rightly weighs biodiversity, landscape and policy arguments that are not addressed here. The hydrological finding is flagged so that it can be weighed explicitly alongside those other considerations.

11. What could be done next

The water balance analysis in this report expresses inputs and losses as water table head change (how far the water table moves each month) rather than as absolute volumes of water. Converting these results into millimetres per year — the more familiar unit for water balance planning — requires a direct measurement of specific yield: how much water is actually stored per metre of water table change in each part of the site. This is achievable through simple field tests at two or three representative wells in each hydrogeological zone, and is recommended as a practical next piece of work.

A second priority is measurement on the ridge itself. The ridge recharge pathway has now been tested both by a boundary-intercept audit of the dipwell data (which confirmed the residual flux) and by a residual-lag analysis (which ruled out a pulse-delivery signature and is consistent with steady-baseflow delivery). What the dune-field dipwells cannot tell us is how much water is actually arriving on the ridge and how much of it is travelling through the ridge aquifer on what timescale. A rain gauge on the ridge crest, a short piezometer transect down the ridge slopes, and ideally a ground-penetrating radar survey of the aquifer thickness beneath the plantation, would together allow the ridge recharge pathway to be modelled explicitly rather than inferred from residuals.

Beyond Newborough Warren, the analytical methods developed in this research — hierarchical clustering, state-space modelling, before-and-after intervention analysis with climate correction, and the summer transfer functions in Section 7 — can in principle be applied to any lowland dune, wetland or shallow aquifer site with a dipwell network and access to a nearby climate station. The data requirements are minimal, the code is publicly available, and records of eight to ten years or more are sufficient for meaningful analysis. Sites across the Atlantic dune zone — from the Sefton Coast to the Gower, Braunton Burrows to Murlough — face comparable challenges.

Summary of key findings

1. The water table at Newborough Warren is declining at approximately 1 to 1.7 cm per year in summer across every part of the site, including clusters on drainage systems independent of the plantation. Winter flooding frequency shows no significant declining trend.
2. The decline is widespread and cannot be explained by the forest alone. The most direct evidence points to rising summer evaporative demand as a site-wide driver. A long-run decline in ridge subsidy cannot be ruled out as a contributing factor, but would require direct measurement on the ridge itself to confirm.
3. The December 2017 clearfell appears to have caused a step lowering of the water table in the impact zone, on top of a prior lowering attributable to the upslope-draining effect of the 2015 scraping at CEH36. The cumulative climate-corrected deficit has continued to grow since felling. The net hydrological effect has been negative for dune slack ecology in the affected area.
4. The plantation reduces rainfall recharge (about 24 per cent of rainfall is intercepted by the canopy), and the water table beneath it is less sensitive to summer evaporative demand than the water table under open dune. This does not mean the forest is hydrologically neutral: interception plus transpiration may still draw the water table deeper than open dune would in the same location, and Forest-adjacent C3 slacks on the same drainage pathway are consistently deeper than interior C3 slacks.
5. Dune scraping can deliver real local benefits — 14 cm of climate-corrected improvement at CEH36. A scraped hollow also acts as a drainage sink for the area upslope of it; the hydrological footprint of scraping therefore extends beyond the scraped slack itself.
6. Summer minimum water table depth — not winter rainfall — is the dominant control on winter flooding in the Western Block. Forecasting equations for summer minima are provided in Section 7.
7. Western dune slacks appear to have already passed the point at which realistic winter rainfall can restore flooding. Eastern slacks remain viable but may cross critical thresholds in the early 2030s.
8. The water balance analysis shows a structured residual flux in every cluster that direct rainfall recharge does not account for. The likely sources differ by cluster: seepage from Llyn Rhos-Ddu at C1 (where the lake intercepts and stores ridge recharge before releasing it gradually), direct ridge recharge at C4 (through the plantation), and a mix of smaller contributions at C2 and C3. Direct measurement at these source candidates is needed to confirm.
9. Evidence from a single monitored broadleaf block suggests broadleaf conversion may deepen summer water tables through a two-phase seasonal mechanism. This is indicative rather than conclusive and warrants further investigation.

Where to find out more

The full technical report, analysis code, data and supporting materials are available at:

https://github.com/newbroman/Newborough_Hydrology

Groundwater monitoring data were collected independently by the author. Climate data are from the RAF Valley Met Office station under the Open Government Licence. Topographic data are from Natural Resources Wales LiDAR under the Open Government Licence v3.0.